

IE7615 Milestone 1 - results summary

Data

- Indexed images: 7343 in 73 folders; junk entries skipped: 2; corrupt: 0
- Background-only images removed: 385 (filename tags *bg* / *BG###*; OBJ004 has no tags, so its background shots stay in)
- Exact duplicates removed from object photos (MD5, after background removal): 16 same-class + 0 cross-class
- Final: 6942 images, 73 classes, 89-106 per class
- Split 70/15/15 stratified, seed 42: train 4859 / val 1041 / test 1042
- Label audit (5-fold out-of-fold): 97.9% ; suspects: 50
- Near-duplicate test images (cos >= 0.92): 99; hard subset: 943
- Class-weighted loss: on (train ratio 1.19)

Note: the pretrained models were shortlisted from earlier experiments (Tanmayi's and Santhosh's runs); the final model is chosen on validation data only.

Models

model	description	params_M	trainable_M	epochs	batch	lr_head	lr_backbone	aug
A2_CustomCNN_w384	Custom CNN (Santhosh): w384, Kaiming, 60 ep, bs32	3.41	3.415	60	32	0.002	0.002	robust
C_ResNet50	ResNet-50 fine-tuned	23.66	23.658	20	32	0.001	0.0001	robust
D_EfficientNetB0	EfficientNet-B0 fine-tuned	4.1	4.101	20	32	0.001	0.0003	robust
E_ConvNeXtT	ConvNeXt-Tiny fine-tuned	27.88	27.876	20	32	0.001	0.0001	robust
G_DenseNet121	DenseNet-121 fine-tuned	7.03	7.029	20	32	0.001	0.0001	robust

Main comparison (test set)

model	val_f1	test_acc	acc_95CI	test_f1	hard_acc	top5_acc	tta_atest_robust_mean	test_errors	params_M	train_min	ms_per_img	
A2_CustomCNN_w384	89.56	89.06	[87.1, 91.1]	88.72	87.91	97.12	91.17	66.96	114	3.41	30.58	0.77

model	val_f1	test_acc	acc_95CI	test_f1	hard_acc	top5_acc	tta_actest_robust_mean	test_errors	params_M	train_min	ms_per_img	
C_ResNet50	99.22	98.66	[98.0, 99.3]	98.65	98.52	99.71	98.85	78.96	14	23.66	11.43	1.26
D_EfficientNetB0	99.41	98.75	[98.1, 99.3]	98.74	98.62	99.81	98.94	81.81	13	4.1	11.71	0.7
E_ConvNeXtT	99.61	99.23	[98.7, 99.7]	99.23	99.15	99.81	99.23	84.74	8	27.88	13.4	1.43
G_DenseNet121	99.13	98.66	[97.9, 99.3]	98.63	98.52	99.9	98.85	73.59	14	7.03	14.09	1.51

Selection (validation only)

model	val_f1	val_robust_mean	params_M
E_ConvNeXtT	99.61	83.98	27.88
D_EfficientNetB0	99.41	81.63	4.1
C_ResNet50	99.22	79.04	23.66
G_DenseNet121	99.13	73.87	7.03

Selected model: E_ConvNeXtT - rule: within 0.5 pt of best val macro-F1 -> highest perturbed-val accuracy -> fewest params.

McNemar exact test vs selected model

compared_with	only_selected_correct	only_other_correct	p_value	significant
A2_CustomCNN_w384	110	4	0	yes
C_ResNet50	11	5	0.2101	no
D_EfficientNetB0	10	5	0.3018	no
G_DenseNet121	9	3	0.146	no

Robustness (test accuracy %)

	clean	small_object	cropped	occluded	low_quality	robust_mean	mean_drop
A2_CustomCNN_w384	89.06	76.68	90.12	74.95	26.1	66.96	22.1
C_ResNet50	98.66	95.2	98.08	89.64	32.92	78.96	19.7

	clean	small_object	cropped	occluded	low_quality	robust_mean	mean_drop
D_EfficientNetB0	98.75	94.82	97.98	89.44	45.01	81.81	16.94
E_ConvNeXtT	99.23	97.6	98.46	91.17	51.73	84.74	14.49
G_DenseNet121	98.66	93.47	98.18	82.82	19.87	73.59	25.07

Background-shortcut test

model	n_bg_images	own_id_predicted_pct	chance_pct	mean_conf_on_bg	mean_conf_on_test_objects
A2_CustomCNN_w384	378	33.069	1.37	0.377	0.799
C_ResNet50	378	36.508	1.37	0.346	0.867
D_EfficientNetB0	378	38.095	1.37	0.376	0.886
E_ConvNeXtT	378	35.714	1.37	0.356	0.89
G_DenseNet121	378	34.656	1.37	0.332	0.867

Mistakes per model

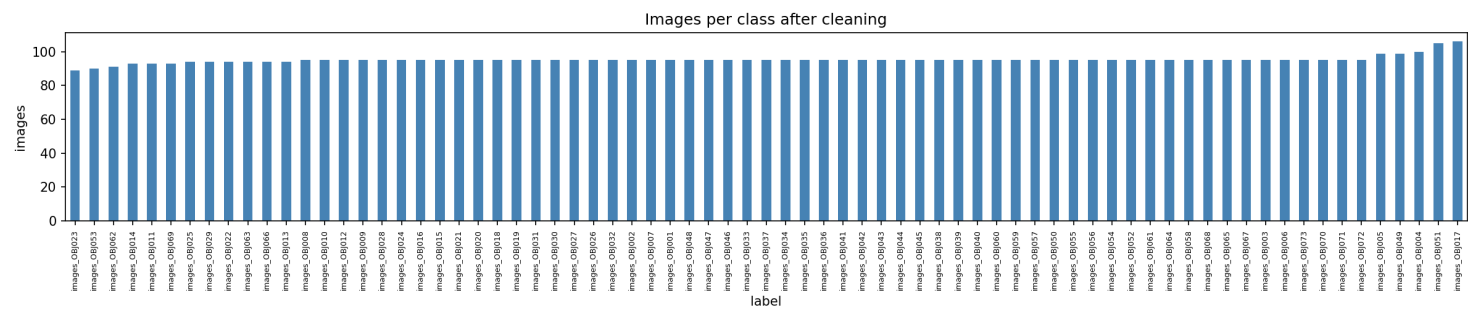
model	test	val
A2_CustomCNN_w384	114	106
C_ResNet50	14	8
D_EfficientNetB0	13	6
E_ConvNeXtT	8	4
G_DenseNet121	14	9

Weakest classes (selected model)

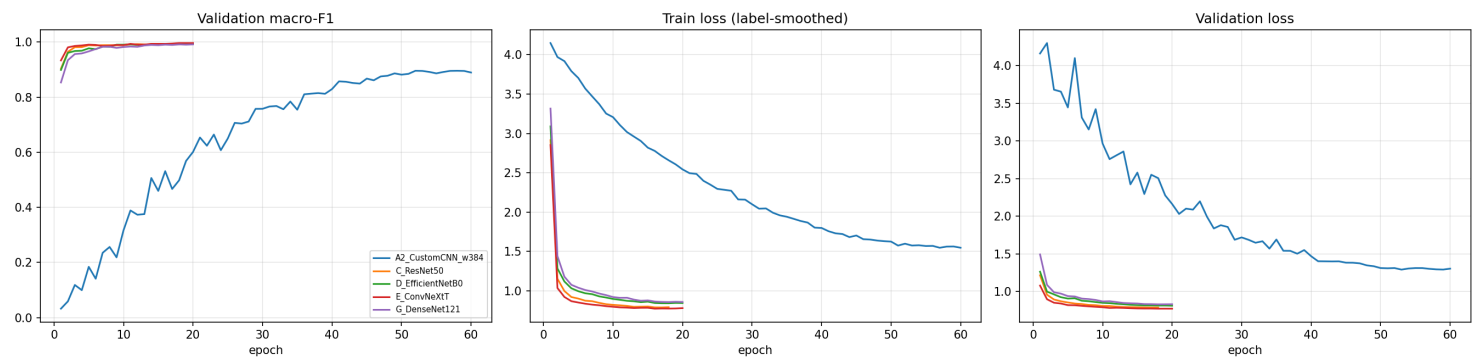
	precision	recall	f1-score	support
images_OBJ061	0.929	0.929	0.929	14
images_OBJ027	0.875	1	0.933	14

	precision	recall	f1-score	support
images_OBJ054	0.933	0.933	0.933	15
images_OBJ014	1	0.929	0.963	14
images_OBJ042	1	0.929	0.963	14
images_OBJ037	1	0.929	0.963	14
images_OBJ012	1	0.929	0.963	14
images_OBJ063	1	0.929	0.963	14
images_OBJ059	1	0.929	0.963	14
images_OBJ045	0.933	1	0.966	14

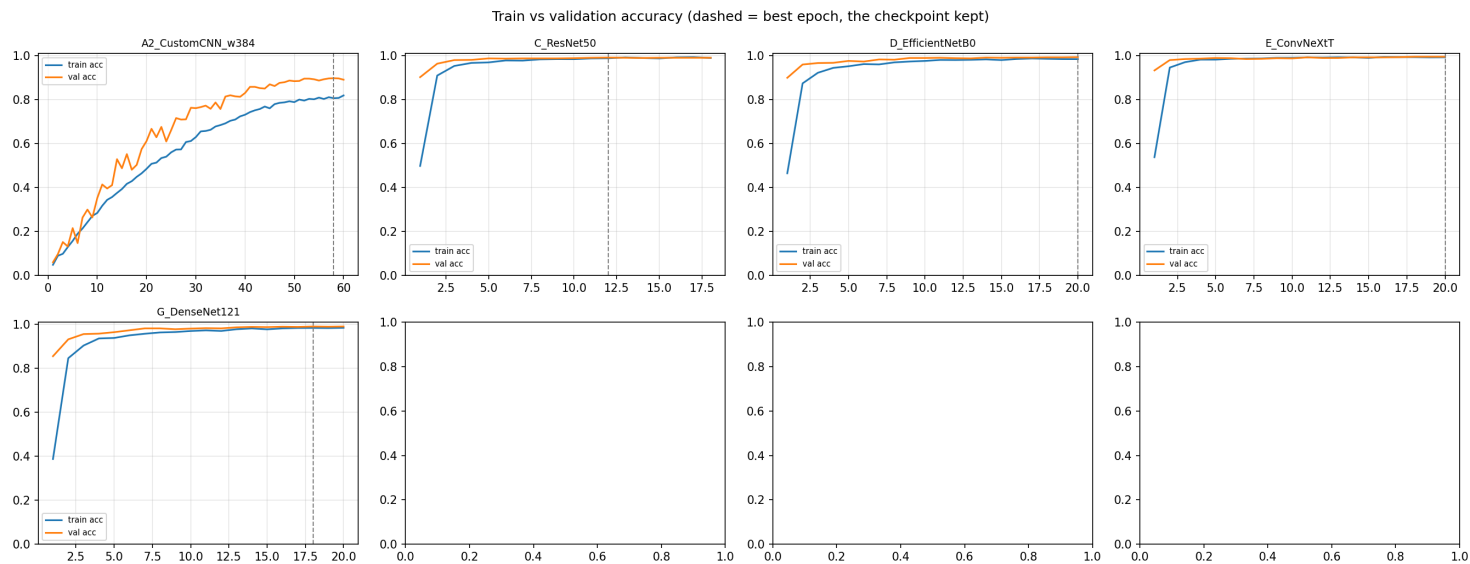
Images per class



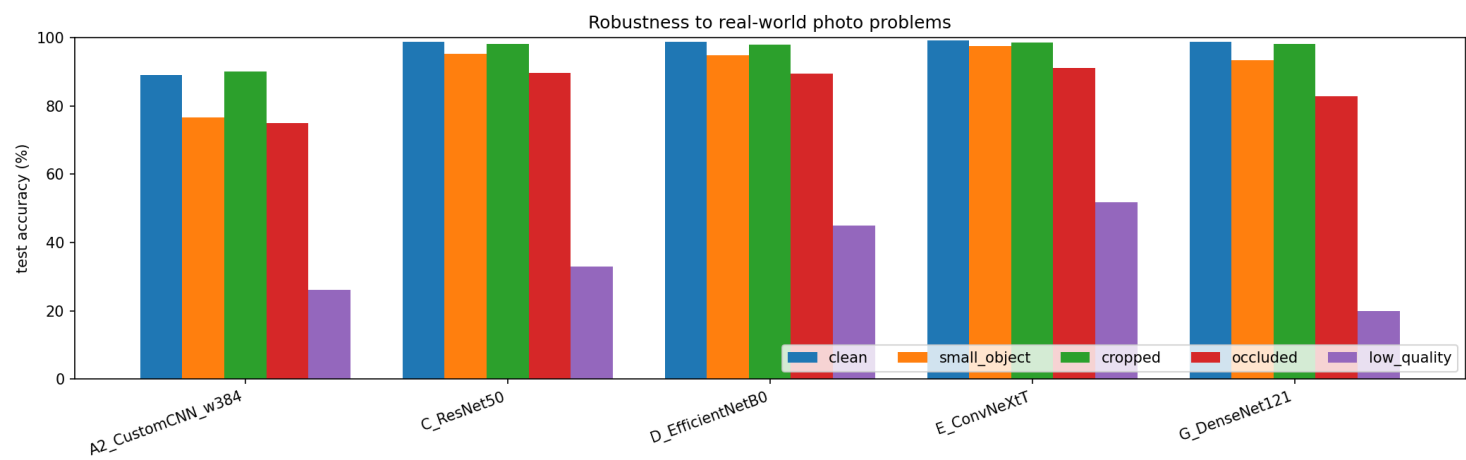
Training curves



Train vs validation accuracy



Robustness



Perturbation examples

clean



small_object



cropped



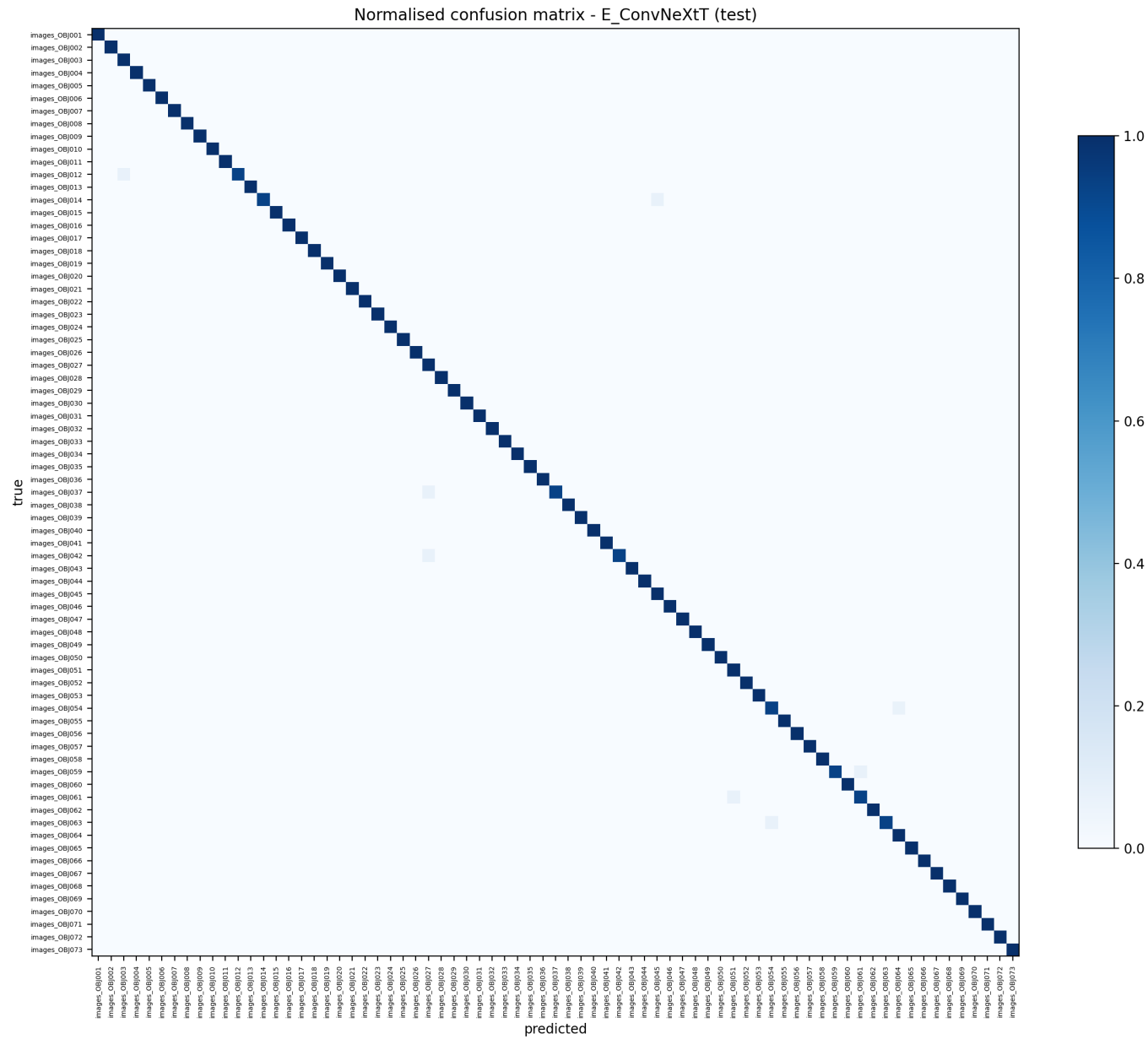
occluded



low_quality



Confusion matrix - E_ConvNeXtT



Test mistakes - E_ConvNeXtT

E_ConvNeXtT - test mistakes (8 total, showing 8 most confident)



Grad-CAM

Grad-CAM for every model (green = correct, red = wrong)

